Name	Vedant Dinkar Patil
UID no.	2024300170
Experiment No.	Exp - 4

AIM:	Single Layer Perceptron implementation
PROBLEM STATEMENT :	Implement AND gate using perceptron learning rule. Display X1,X2,t,Y, W1,W2,B at end of epoch1 and epoch2 YIN= B+SUM(WiXi) Y IS CALCULATED AFTER APPLYING ACTIVATION FUNCTION HERE Y=0 IF YIN=0, ELSE Y= -1 IF YIN < 0, ELSE Y = 1 IF YIN>0 DELTA W1 = ALPHA*t*X1 DELTA W2 = ALPHA*t*X2 DELTA B = ALPHA*t ie W1 new= W1old + delta W1 ie W2 new= W2old + delta W2 $B_{NEW}=B_{OLD}$+ALPHA*t IF t = Y THEN NO CHANGE IN WEIGHTS, WEIGHTS REMAINS SAME
Theory:	A Single-Layer Perceptron (SLP) is a foundational neural network unit that classifies linearly s passing it through a bipolar signum activation function ($y = 1$ if $y_{in} > 0$, 0 if $y_{in} = 0$, and -1 perceptron learning rule $\Delta w_i = \alpha t x_i$ and $\Delta b = \alpha t$ (with learning rate $\alpha = 1$). Applying this ru $[1,1]$ and $t = -1$ otherwise) starting from zero initial weights and bias ($w_1 = 0, w_2 = 0, b = 0$ $y_{in} = 0 \rightarrow y = 0$, updating weights to $w_1 = 1, w_2 = 1, b = 1$; Pattern 2 ($[1, -1], t = -1$) yield ($[-1,1], t = -1$) yields $y_{in} = 2 \rightarrow y = 1$, updating to $w_1 = 1, w_2 = 1, b = -1$; and Pattern 4 (**Epoch 1** with parameters $w_1 = 1, w_2 = 1, b = -1$ and step outputs $[y_{in}, y]$ across the four patt these weights gives $y_{in} = 1 \rightarrow y = 1$ for Pattern 1, $y_{in} = -1 \rightarrow y = -1$ for Pattern 2, $y_{in} = -$

	calculated output y precisely matches its target t, no weight adjustments occur ($\Delta w_1 = \Delta w_2 =$ parameters $w_1 = 1, w_2 = 1, b = -1$.

PROGRAM:	Perceptron_AND_Gate::
	#include <iostream>
	#include <iomanip>
	
	using namespace std;
	
	// 1. FIXED: Activation function now returns binary (0 or 1) to match targets
	int activation(double yin) {
	  if (yin > 0) return 1;
	  else return 0;
	}
	
	int main() {
	  double X1[] = {0, 0, 1, 1};
	  double X2[] = {0, 1, 0, 1};
	  double t[] = {0, 0, 0, 1}; // AND gate
	
	  double W1 = 0.0;
	  double W2 = 0.0;
	  double B = 0.0;
	  double alpha;
	
	  int num_samples = 4;
	  int max_epochs;
	
	  cout << "Enter learning rate (alpha): ";
	  cin >> alpha;
	  cout << "Enter maximum number of epochs: ";
	  cin >> max_epochs;
	
	  bool converged = false;
	
	  for (int epoch = 1; epoch <= max_epochs; ++epoch) {
	  cout <<
	"\n==========
	  cout << " EPOCH " << epoch << "

```
        cout <<
"=============================================================================
        cout << left << setw(8) << "Sample"
             << setw(6) << "X1"
             << setw(6) << "X2"
             << setw(7) << "W1"
             << setw(7) << "W2"
             << setw(6) << "B"
             << setw(6) << "Target"
             << setw(8) << "Yin"
             << setw(5) << "Y"
             << setw(8) << "dW1"
             << setw(8) << "dW2"
             << setw(6) << "dB"
             << setw(8) << "NewW1"
             << setw(8) << "NewW2"
             << setw(6) << "NewB" << "\n";
        cout << "-----------------------------------------------------------------------------------------------------

        bool weight_updated = false;

        for (int i = 0; i < num_samples; ++i) {
            double current_w1 = W1;
            double current_w2 = W2;
            double current_b  = B;

            double yin = current_b + (current_w1 * X1[i]) + (current_w2 * X2[i]);
            int y = activation(yin);

            double delta_w1 = 0.0;
            double delta_w2 = 0.0;
            double delta_b  = 0.0;

            if (t[i] != y) {
                // 2. FIXED: Use (Target - Predicted Error) for the update rule
                double error = t[i] - y;
                delta_w1 = alpha * error * X1[i];
                delta_w2 = alpha * error * X2[i];
                delta_b  = alpha * error;
```

```
            W1 = current_w1 + delta_w1;
            W2 = current_w2 + delta_w2;
            B  = current_b + delta_b;

            weight_updated = true;
        }

        cout << left << setw(8) << (i + 1)
             << setw(6) << X1[i]
             << setw(6) << X2[i]
             << setw(7) << fixed << setprecision(1) << current_w1
             << setw(7) << current_w2
             << setw(6) << current_b
             << setw(6) << t[i]
             << setw(8) << yin
             << setw(5) << y
             << setw(8) << delta_w1
             << setw(8) << delta_w2
             << setw(6) << delta_b
             << setw(8) << W1
             << setw(8) << W2
             << setw(6) << B << "\n";
    }

    cout << "----------------------------------------------------------------------------------------------------

    // If no weights were updated in this epoch, we found the solution!
    if (!weight_updated) {
        converged = true;
        cout << "\nSUCCESS: Solution found and converged at Epoch " << epoch << "!\n";
        break;
    }
  }

  if (!converged) {
    cout << "\nFAILURE: Did not find a solution. Epoch limit reached.\n";
  }

  return 0;
}
```

Perceptron_OR_Gate::

```
#include <iostream>
#include <iomanip>

using namespace std;

// Activation function returns binary (0 or 1) to match targets
int activation(double yin) {
    if (yin > 0) return 1;
    else return 0;
}

int main() {
    double X1[] = {0, 0, 1, 1};
    double X2[] = {0, 1, 0, 1};

    // CHANGED: Targets updated for the OR logic gate
    double t[]  = {0, 1, 1, 1};

    double W1 = 0.0;
    double W2 = 0.0;
    double B = 0.0;
    double alpha;

    int num_samples = 4;
    int max_epochs;

    cout << "Enter learning rate (alpha): ";
    cin >> alpha;
    cout << "Enter maximum number of epochs: ";
    cin >> max_epochs;

    bool converged = false;

    for (int epoch = 1; epoch <= max_epochs; ++epoch) {
        cout <<
"\n==========================================================================
        cout << "                                        EPOCH " << epoch << "
        cout <<
"==========================================================================
```

```
    cout << left << setw(8) << "Sample"
         << setw(6) << "X1"
         << setw(6) << "X2"
         << setw(7) << "W1"
         << setw(7) << "W2"
         << setw(6) << "B"
         << setw(6) << "Target"
         << setw(8) << "Yin"
         << setw(5) << "Y"
         << setw(8) << "dW1"
         << setw(8) << "dW2"
         << setw(6) << "dB"
         << setw(8) << "NewW1"
         << setw(8) << "NewW2"
         << setw(6) << "NewB" << "\n";
    cout << "----------------------------------------------------------------------------------------------------

    bool weight_updated = false;

    for (int i = 0; i < num_samples; ++i) {
       double current_w1 = W1;
       double current_w2 = W2;
       double current_b  = B;

       double yin = current_b + (current_w1 * X1[i]) + (current_w2 * X2[i]);
       int y = activation(yin);

       double delta_w1 = 0.0;
       double delta_w2 = 0.0;
       double delta_b  = 0.0;

       if (t[i] != y) {
          // Use (Target - Predicted Error) for the update rule
          double error = t[i] - y;
          delta_w1 = alpha * error * X1[i];
          delta_w2 = alpha * error * X2[i];
          delta_b  = alpha * error;

          W1 = current_w1 + delta_w1;
          W2 = current_w2 + delta_w2;
```

```
            B  = current_b + delta_b;

            weight_updated = true;
        }

        cout << left << setw(8) << (i + 1)
            << setw(6) << X1[i]
            << setw(6) << X2[i]
            << setw(7) << fixed << setprecision(1) << current_w1
            << setw(7) << current_w2
            << setw(6) << current_b
            << setw(6) << t[i]
            << setw(8) << yin
            << setw(5) << y
            << setw(8) << delta_w1
            << setw(8) << delta_w2
            << setw(6) << delta_b
            << setw(8) << W1
            << setw(8) << W2
            << setw(6) << B << "\n";
    }

    cout << "----------------------------------------------------------------------------------------------------

    // If no weights were updated in this epoch, we found the solution!
    if (!weight_updated) {
        converged = true;
        cout << "\nSUCCESS: Solution found and converged at Epoch " << epoch << "!\n";
        break;
    }
  }

  if (!converged) {
    cout << "\nFAILURE: Did not find a solution. Epoch limit reached.\n";
  }

  return 0;
}
```

Output:

AND_GATE_OUTPUT

```
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$ g++ perceptron.cpp -o perceptron
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$ ./perceptron
Enter learning rate (alpha): 1
Enter maximum number of epochs: 10

================================================================================================================
                                                EPOCH 1
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0     0     0.0   0.0   0.0   0.0   0.0     0     0.0     0.0     0.0   0.0     0.0     0.0
2       0.0   1.0   0.0   0.0   0.0   0.0   0.0     0     0.0     0.0     0.0   0.0     0.0     0.0
3       1.0   0.0   0.0   0.0   0.0   0.0   0.0     0     0.0     0.0     0.0   0.0     0.0     0.0
4       1.0   1.0   0.0   0.0   0.0   1.0   0.0     0     1.0     1.0     1.0   1.0     1.0     1.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 2
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   1.0   1.0   1.0   0.0   1.0     1     -0.0    -0.0    -1.0  1.0     1.0     0.0
2       0.0   1.0   1.0   1.0   0.0   0.0   1.0     1     -0.0    -1.0    -1.0  1.0     0.0     -1.0
3       1.0   0.0   1.0   0.0   -1.0  0.0   0.0     0     0.0     0.0     0.0   1.0     0.0     -1.0
4       1.0   1.0   1.0   0.0   -1.0  1.0   0.0     0     1.0     1.0     1.0   2.0     1.0     0.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 3
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   2.0   1.0   0.0   0.0   0.0     0     0.0     0.0     0.0   2.0     1.0     0.0
2       0.0   1.0   2.0   1.0   0.0   0.0   1.0     1     -0.0    -1.0    -1.0  2.0     0.0     -1.0
3       1.0   0.0   2.0   0.0   -1.0  0.0   1.0     1     -1.0    -0.0    -1.0  1.0     0.0     -2.0
4       1.0   1.0   1.0   0.0   -2.0  1.0   -1.0    0     1.0     1.0     1.0   2.0     1.0     -1.0
----------------------------------------------------------------------------------------------------------------

Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   2.0   1.0   -1.0  0.0   -1.0    0     0.0     0.0     0.0   2.0     1.0     -1.0
2       0.0   1.0   2.0   1.0   -1.0  0.0   0.0     0     0.0     0.0     0.0   2.0     1.0     -1.0
3       1.0   0.0   2.0   1.0   -1.0  0.0   1.0     1     -1.0    -0.0    -1.0  1.0     1.0     -2.0
4       1.0   1.0   1.0   1.0   -2.0  1.0   0.0     0     1.0     1.0     1.0   2.0     2.0     -1.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 5
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   2.0   2.0   -1.0  0.0   -1.0    0     0.0     0.0     0.0   2.0     2.0     -1.0
2       0.0   1.0   2.0   2.0   -1.0  0.0   1.0     1     -0.0    -1.0    -1.0  2.0     1.0     -2.0
3       1.0   0.0   2.0   1.0   -2.0  0.0   0.0     0     0.0     0.0     0.0   2.0     1.0     -2.0
4       1.0   1.0   2.0   1.0   -2.0  1.0   1.0     1     0.0     0.0     0.0   2.0     1.0     -2.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 6
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   2.0   1.0   -2.0  0.0   -2.0    0     0.0     0.0     0.0   2.0     1.0     -2.0
2       0.0   1.0   2.0   1.0   -2.0  0.0   -1.0    0     0.0     0.0     0.0   2.0     1.0     -2.0
3       1.0   0.0   2.0   1.0   -2.0  0.0   0.0     0     0.0     0.0     0.0   2.0     1.0     -2.0
4       1.0   1.0   2.0   1.0   -2.0  1.0   1.0     1     0.0     0.0     0.0   2.0     1.0     -2.0
----------------------------------------------------------------------------------------------------------------

SUCCESS: Solution found and converged at Epoch 6!
```

```
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$ ./perceptron
Enter learning rate (alpha): -1
Enter maximum number of epochs: 40

================================================================================================================
                                                EPOCH 1
================================================================================================================
Sample  X1    X2    W1     W2     B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0     0     0.0    0.0    0.0   0.0   0.0     0    0.0     0.0     0.0   0.0     0.0     0.0
2       0.0   1.0   0.0    0.0    0.0   0.0   0.0     0    0.0     0.0     0.0   0.0     0.0     0.0
3       1.0   0.0   0.0    0.0    0.0   0.0   0.0     0    0.0     0.0     0.0   0.0     0.0     0.0
4       1.0   1.0   0.0    0.0    0.0   1.0   0.0     0    -1.0    -1.0    -1.0  -1.0    -1.0    -1.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 2
================================================================================================================
Sample  X1    X2    W1     W2     B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   -1.0   -1.0   -1.0  0.0   -1.0    0    0.0     0.0     0.0   -1.0    -1.0    -1.0
2       0.0   1.0   -1.0   -1.0   -1.0  0.0   -2.0    0    0.0     0.0     0.0   -1.0    -1.0    -1.0
3       1.0   0.0   -1.0   -1.0   -1.0  0.0   -2.0    0    0.0     0.0     0.0   -1.0    -1.0    -1.0
4       1.0   1.0   -1.0   -1.0   -1.0  1.0   -3.0    0    -1.0    -1.0    -1.0  -2.0    -2.0    -2.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 3
================================================================================================================
Sample  X1    X2    W1     W2     B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   -2.0   -2.0   -2.0  0.0   -2.0    0    0.0     0.0     0.0   -2.0    -2.0    -2.0
2       0.0   1.0   -2.0   -2.0   -2.0  0.0   -4.0    0    0.0     0.0     0.0   -2.0    -2.0    -2.0
3       1.0   0.0   -2.0   -2.0   -2.0  0.0   -4.0    0    0.0     0.0     0.0   -2.0    -2.0    -2.0
4       1.0   1.0   -2.0   -2.0   -2.0  1.0   -6.0    0    -1.0    -1.0    -1.0  -3.0    -3.0    -3.0
----------------------------------------------------------------------------------------------------------------
```

```
================================================================================================================
                                                EPOCH 39
================================================================================================================
Sample  X1    X2    W1     W2     B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   -38.0  -38.0  -38.0 0.0   -38.0   0    0.0     0.0     0.0   -38.0   -38.0   -38.0
2       0.0   1.0   -38.0  -38.0  -38.0 0.0   -76.0   0    0.0     0.0     0.0   -38.0   -38.0   -38.0
3       1.0   0.0   -38.0  -38.0  -38.0 0.0   -76.0   0    0.0     0.0     0.0   -38.0   -38.0   -38.0
4       1.0   1.0   -38.0  -38.0  -38.0 1.0   -114.0  0    -1.0    -1.0    -1.0  -39.0   -39.0   -39.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 40
================================================================================================================
Sample  X1    X2    W1     W2     B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   -39.0  -39.0  -39.0 0.0   -39.0   0    0.0     0.0     0.0   -39.0   -39.0   -39.0
2       0.0   1.0   -39.0  -39.0  -39.0 0.0   -78.0   0    0.0     0.0     0.0   -39.0   -39.0   -39.0
3       1.0   0.0   -39.0  -39.0  -39.0 0.0   -78.0   0    0.0     0.0     0.0   -39.0   -39.0   -39.0
4       1.0   1.0   -39.0  -39.0  -39.0 1.0   -117.0  0    -1.0    -1.0    -1.0  -40.0   -40.0   -40.0
----------------------------------------------------------------------------------------------------------------

FAILURE: Did not find a solution. Epoch limit reached.
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$
```

OR_GATE_OUTPUT

```
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$ g++ perceptron_or.cpp -o perceptron_or
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$ ./perceptron_or
Enter learning rate (alpha): 1
Enter maximum number of epochs: 10

================================================================================================================
                                                EPOCH 1
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0     0     0.0   0.0   0.0   0.0   0.0     0     0.0     0.0     0.0    0.0     0.0     0.0
2       0.0   1.0   0.0   0.0   0.0   1.0   0.0     0     0.0     1.0     1.0    0.0     1.0     1.0
3       1.0   0.0   0.0   1.0   1.0   1.0   1.0     1     0.0     0.0     0.0    0.0     1.0     1.0
4       1.0   1.0   0.0   1.0   1.0   1.0   2.0     1     0.0     0.0     0.0    0.0     1.0     1.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 2
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   0.0   1.0   1.0   0.0   1.0     1     -0.0    -0.0    -1.0   0.0     1.0     0.0
2       0.0   1.0   0.0   1.0   0.0   1.0   1.0     1     0.0     0.0     0.0    0.0     1.0     0.0
3       1.0   0.0   0.0   1.0   0.0   1.0   0.0     0     1.0     0.0     1.0    1.0     1.0     1.0
4       1.0   1.0   1.0   1.0   1.0   1.0   3.0     1     0.0     0.0     0.0    1.0     1.0     1.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 3
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   1.0   1.0   1.0   0.0   1.0     1     -0.0    -0.0    -1.0   1.0     1.0     0.0
2       0.0   1.0   1.0   1.0   0.0   1.0   1.0     1     0.0     0.0     0.0    1.0     1.0     0.0
3       1.0   0.0   1.0   1.0   0.0   1.0   1.0     1     0.0     0.0     0.0    1.0     1.0     0.0
4       1.0   1.0   1.0   1.0   0.0   1.0   2.0     1     0.0     0.0     0.0    1.0     1.0     0.0
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 4
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   1.0   1.0   0.0   0.0   0.0     0     0.0     0.0     0.0    1.0     1.0     0.0
2       0.0   1.0   1.0   1.0   0.0   1.0   1.0     1     0.0     0.0     0.0    1.0     1.0     0.0
3       1.0   0.0   1.0   1.0   0.0   1.0   1.0     1     0.0     0.0     0.0    1.0     1.0     0.0
4       1.0   1.0   1.0   1.0   0.0   1.0   2.0     1     0.0     0.0     0.0    1.0     1.0     0.0
----------------------------------------------------------------------------------------------------------------

SUCCESS: Solution found and converged at Epoch 4!
```

```
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$ ./perceptron_or
Enter learning rate (alpha): -0.1
Enter maximum number of epochs: 30

================================================================================================================
                                                EPOCH 1
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0     0     0.0   0.0   0.0   0.0   0.0     0     0.0     0.0     0.0    0.0     0.0     0.0
2       0.0   1.0   0.0   0.0   0.0   1.0   0.0     0     -0.0    -0.1    -0.1   0.0     -0.1    -0.1
3       1.0   0.0   0.0   -0.1  -0.1  1.0   -0.1    0     -0.1    -0.0    -0.1   -0.1    -0.1    -0.2
4       1.0   1.0   -0.1  -0.1  -0.2  1.0   -0.4    0     -0.1    -0.1    -0.1   -0.2    -0.2    -0.3
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 2
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   -0.2  -0.2  -0.3  0.0   -0.3    0     0.0     0.0     0.0    -0.2    -0.2    -0.3
2       0.0   1.0   -0.2  -0.2  -0.3  1.0   -0.5    0     -0.0    -0.1    -0.1   -0.2    -0.3    -0.4
3       1.0   0.0   -0.2  -0.3  -0.4  1.0   -0.6    0     -0.1    -0.0    -0.1   -0.3    -0.3    -0.5
4       1.0   1.0   -0.3  -0.3  -0.5  1.0   -1.1    0     -0.1    -0.1    -0.1   -0.4    -0.4    -0.6
----------------------------------------------------------------------------------------------------------------

================================================================================================================
                                                EPOCH 3
================================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y     dW1     dW2     dB     NewW1   NewW2   NewB
----------------------------------------------------------------------------------------------------------------
1       0.0   0.0   -0.4  -0.4  -0.6  0.0   -0.6    0     0.0     0.0     0.0    -0.4    -0.4    -0.6
2       0.0   1.0   -0.4  -0.4  -0.6  1.0   -1.0    0     -0.0    -0.1    -0.1   -0.4    -0.5    -0.7
3       1.0   0.0   -0.4  -0.5  -0.7  1.0   -1.1    0     -0.1    -0.0    -0.1   -0.5    -0.5    -0.8
4       1.0   1.0   -0.5  -0.5  -0.8  1.0   -1.8    0     -0.1    -0.1    -0.1   -0.6    -0.6    -0.9
----------------------------------------------------------------------------------------------------------------
```

```
================================================================================================
                                            EPOCH 28
================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
------------------------------------------------------------------------------------------------
1       0.0   0.0   -5.4  -5.4  -8.1  0.0   -8.1    0    0.0     0.0     0.0   -5.4    -5.4    -8.1
2       0.0   1.0   -5.4  -5.4  -8.1  1.0   -13.5   0    -0.0    -0.1    -0.1  -5.4    -5.5    -8.2
3       1.0   0.0   -5.4  -5.5  -8.2  1.0   -13.6   0    -0.1    -0.0    -0.1  -5.5    -5.5    -8.3
4       1.0   1.0   -5.5  -5.5  -8.3  1.0   -19.3   0    -0.1    -0.1    -0.1  -5.6    -5.6    -8.4
------------------------------------------------------------------------------------------------

================================================================================================
                                            EPOCH 29
================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
------------------------------------------------------------------------------------------------
1       0.0   0.0   -5.6  -5.6  -8.4  0.0   -8.4    0    0.0     0.0     0.0   -5.6    -5.6    -8.4
2       0.0   1.0   -5.6  -5.6  -8.4  1.0   -14.0   0    -0.0    -0.1    -0.1  -5.6    -5.7    -8.5
3       1.0   0.0   -5.6  -5.7  -8.5  1.0   -14.1   0    -0.1    -0.0    -0.1  -5.7    -5.7    -8.6
4       1.0   1.0   -5.7  -5.7  -8.6  1.0   -20.0   0    -0.1    -0.1    -0.1  -5.8    -5.8    -8.7
------------------------------------------------------------------------------------------------

================================================================================================
                                            EPOCH 30
================================================================================================
Sample  X1    X2    W1    W2    B     TargetYin     Y    dW1     dW2     dB    NewW1   NewW2   NewB
------------------------------------------------------------------------------------------------
1       0.0   0.0   -5.8  -5.8  -8.7  0.0   -8.7    0    0.0     0.0     0.0   -5.8    -5.8    -8.7
2       0.0   1.0   -5.8  -5.8  -8.7  1.0   -14.5   0    -0.0    -0.1    -0.1  -5.8    -5.9    -8.8
3       1.0   0.0   -5.8  -5.9  -8.8  1.0   -14.6   0    -0.1    -0.0    -0.1  -5.9    -5.9    -8.9
4       1.0   1.0   -5.9  -5.9  -8.9  1.0   -20.7   0    -0.1    -0.1    -0.1  -6.0    -6.0    -9.0
------------------------------------------------------------------------------------------------

FAILURE: Did not find a solution. Epoch limit reached.
students@students-HP-280-G3-SFF-Business-PC:~/Desktop/2024300170_Vedant$
```

CONCLUSION:

In conclusion, the experiment demonstrates that a Single-Layer Perceptron successfully solves
Starting from zero initial conditions, the model required only one full pass through the training
parameters $w_1 = 1$, $w_2 = 1$, and $b = -1$. Epoch 2 served as definitive verification of network
proving the algorithm's mathematical efficiency in adapting weights to satisfy bipolar logic cor